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 Studierichting: Informatica
 Oefeningenreeks 3D nr. 6

$$\begin{aligned}
 f(x) &= \frac{2-3x}{x^2-3x+2} \\
 f'(x) &= \frac{-3(x^2-3x+2) - ((2-3x)(2x-3))}{(x^2-3x+2)^2} \\
 &= \frac{-3x^2+9x-6-4x+6+6x^2-9x}{(x^2-3x+2)^2} \\
 &= \frac{x(3x-4)}{((x-2)(x-1))^2} \\
 f'(x) = 0 &\Leftrightarrow \frac{x(3x-4)}{((x-2)(x-1))^2} = 0
 \end{aligned}$$

x	$-\infty$	0	1	$\frac{4}{3}$	2	$+\infty$			
$f'(x)$	+	0	-	$ ^{(2)}$	-	0	+	$ ^{(2)}$	+
$f(x)$	$\nearrow$	M	$\searrow$	$ $	$\searrow$	m	$\nearrow$	$ $	$\nearrow$
		1		9					

$$y = ax + b$$

$$\left. \begin{array}{l} P(0, 1) \in y \\ Q\left(\frac{4}{3}, 9\right) \in y \end{array} \right\} \Leftrightarrow a = \frac{9-1}{\frac{4}{3}} = 6$$

$$P \text{ invullen in } y = 6x + b$$

$$1 = 6 \cdot 0 + b$$

$$\Leftrightarrow b = 1$$

$$\Rightarrow y = 6x + 1$$

References